FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Main)-2024

TEST DATE: 24-03-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. D

Sol. $\frac{dx}{dt} = 3t - 7$

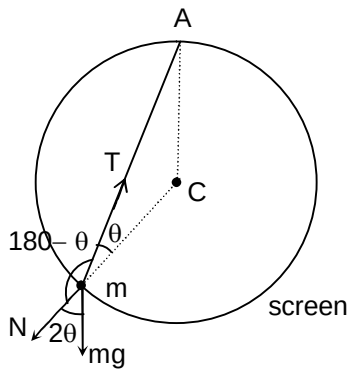
2. C

Sol. FBD of bead

Using sin law,

$\frac{T}{\sin 2\theta} = \frac{mg}{\sin(180° - \theta)}$

$\Rightarrow T = 2mg\cos\theta$



3. B

Sol. FBD of block

$N = k \times \ell = 540$ Newton

$\mu \times N = 54$ Newton

Also $mg = 5.4 \times 10 = 54$ Newton

4. B

Sol. $y = \frac{x^2}{9}$

$\frac{dy}{dx} = \frac{2x}{9}$

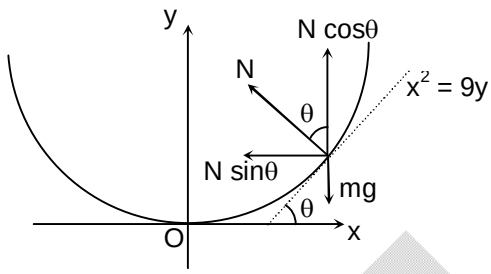
$N\cos\theta = mg$

$N\sin\theta = mx\omega^2$

$\tan\theta = \frac{x\omega^2}{g}$

$\frac{2x}{9} = \frac{x\omega^2}{g}$

$\omega = \sqrt{\frac{2g}{9}}$



5. D

Sol. $v = \sqrt{(10)^2 - 2\times10\times2} = \sqrt{60}$

$a_{c_1} = \frac{v_1^2}{r}, a_{c_2} = \frac{v_2^2}{r}$

$\Delta a_c = \sqrt{a_{c1}{}^2 + a_{c2}{}^2}$

$= 10\sqrt{34}$

6. A

Sol. $P_i = \sqrt{2mE}$

$P_i = P_f$

$\sqrt{2mE} = m(1+2n)v$

$v = \frac{\sqrt{2mE}}{m(2n+1)}$

$(P.E.)_{max} = E - \frac{1}{2}(m+2nm)v^2 = \frac{2nE}{2n+1}$

7. D

Sol. Time of flight does not change because velocity in y direction remains unchanged and time of flight depends only on y component of velocity.

$-2 = 10t - \frac{1}{2}10t^2$

Time of flight $t = 1 + \sqrt{1.4}$ sec ≈ 2.18 sec

Time to reach the wall (t_1)

$t_1 = \frac{4}{10\cos\theta} = 0.4$ sec

Time remaining

$t_2 = 1.78$ sec

Distance covered backward

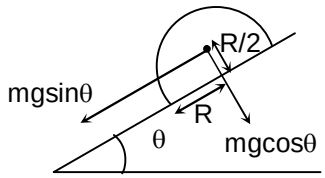
$x = 10 \times 1.78 = 17.8$ m

Distance from thrower = 13.8 m

8. C

Sol. For toppling $(mg \sin\theta)\ \frac{R}{2} > (mg\cos\theta)R$

$\tan\theta > 2$



9. C

Sol. $2m \times g \times \ell \sin\theta = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{2}\frac{2}{3}mR^2\omega^2$

$\left(\omega = \frac{v}{R}\right)$

$v = \sqrt{\frac{3}{2}g\ell\sin\theta}$

10. B

Sol. $x = 8(3\sin\omega t - 4\sin^2\omega t) = 8\sin 3\omega t$

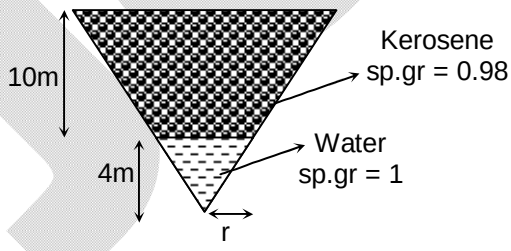
$a_{max} = (3\omega)^2 \times 8 = 72\omega^2$

11. C

Sol. $\frac{r}{4} = \tan 37° = \frac{3}{4}$

r = 3 m

$F = (P_0 + h\rho g)\pi r^2 = 5.6 \times 10^6$ N



12. D

Sol. ΔQ_{av} = 30×1×80 = 2400 cal.

ΔQ_{req} to melt = 50×80 = 4000cal.

Not complete melting

m × 80 = 2400

⇒ 30gm ice melts

250 + 60 = 310gm water +20gm ice.

13. B

Sol. The rate of heat loss by a thin hollow sphere of thickness 'Δx', mean radius 'r' and mode of density 'ρ' is given by :

$mS\frac{dT}{dt} = -\varepsilon\sigma A(T^4 - T_0^4)$

$(\rho 4\pi r^2\Delta x)S\frac{dT}{dt} = -\varepsilon\sigma 4\pi r^2(T^4 - T_0^4)$

$\Rightarrow \frac{dT}{dt} = -\frac{\varepsilon\sigma(T^4 - T_0^4)}{S\Delta x\rho}$

14. B

Sol. $U_E = \frac{1}{2}\varepsilon_0 E^2, E = \frac{\rho r}{3\in_0}$

$$U_E = \frac{\rho^2 r^2}{18 \in_0}$$

15. A

Sol. Since, $T^2 = kr^3$

Differentiating the above equation

$$\Rightarrow 2\frac{\Delta T}{T} = 3\frac{\Delta r}{r}$$

$$\Rightarrow \frac{\Delta T}{T} = \frac{3}{2}\frac{\Delta r}{r}$$

16. A

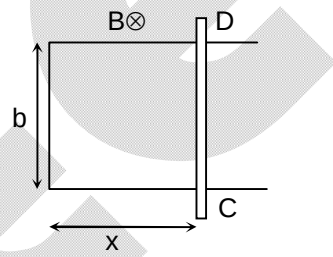
Sol. The magnetic flux must remain constant.

$$\therefore \ \phi_m = B_0 ab = \frac{B_0}{1+kt} bx$$

where x is as shown

$\therefore$ x = a (1+ kt)

Or $v = \frac{dx}{dt} = ak$



17. C

Sol. Use basic concept

18. A

Sol. $$\frac{dv}{dt} = -\left(\frac{f}{u-f}\right)^2 \frac{du}{dt}$$

find $\frac{d^2v}{dt^2}$

19. B

Sol. Use basic concept

20. D

Sol. $$R = \frac{2.5 - 0.5}{5 \times 10^{-3}} = 400\Omega$$

SECTION – B

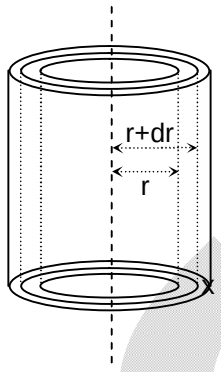
21. 5

Sol. $1 \times \sin 60° = (3\sqrt{3} - y)\sin 90°$

$$y = \frac{5\sqrt{3}}{2}$$

22. 6

Sol. $dm_A = k\left(\frac{r}{R}\right)^2 2\pi r\ell dr$



$dI_A = dm_A\, r^2 = k\frac{r^2}{R^2}2\pi r\ell dr\, r^2$

$I_A = \frac{2\pi k\ell}{R^2}\int_0^R r^5 dr = \frac{2\pi k\ell}{R^2}\frac{R^6}{6} = \frac{2\pi k\ell R^4}{6}$

$dm_B = k\left(\frac{r}{R}\right)^3 2\pi r\ell dr$

$dI_B = dm_B\, r^2 = k\frac{r^3}{R^3}2\pi r\ell dr\, r^2$

$I_B = \frac{2\pi k\ell}{R^3}\int_0^R r^6 dr = \frac{2\pi k\ell}{R^3}\frac{R^7}{7} = \frac{2\pi k\ell R^4}{7}$

$\Rightarrow \frac{I_A}{I_B} = \frac{7}{6}$

23. 4

Sol. Consider two thin discs of width dx each at section ab and cd.

$dR_1 = \frac{\rho dx}{A}$

$P_1 = i^2 dR_1$

$\frac{P_1}{Adx} = \frac{i^2\rho}{A^2}$

$dR_2 = \frac{\rho dx}{A}$

$P_2 = i^2 dR_2$

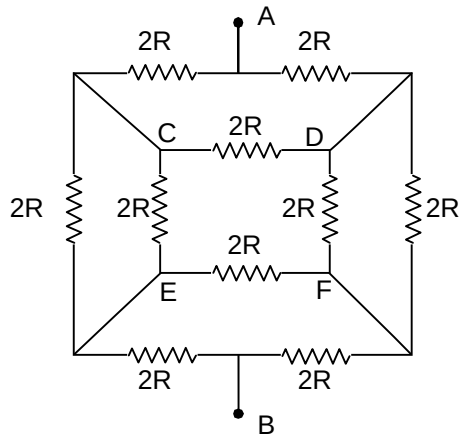
$\frac{P_2}{Adx} = \frac{i^2\rho}{16A^2}$

$P_1 : P_2 = 16 : 1$

24. 5

Sol. No current in CD and EF

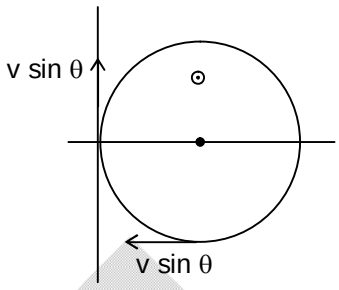
$R_{AB} = \frac{5R}{2}$



25. 2

Sol. Component of velocity parallel to magnetic field will not change

$$|\Delta\vec{v}| = \sqrt{2}v\sin\theta = \frac{v}{\sqrt{2}}$$



26. 2

Sol. $2\pi r = 2\pi r_0 - 2vt$

$$r = r_0 - \frac{vt}{\pi} \Rightarrow \frac{dr}{dt} = -\frac{v}{\pi}$$

$A = \pi r^2$

$$\varepsilon = -\frac{d\phi}{dt} \Rightarrow \varepsilon = -B\pi 2r\left(\frac{dr}{dt}\right)$$

$$i = \varepsilon / R = \frac{1}{\lambda 2\pi r}B\pi 2r\left(\frac{v}{\pi}\right)$$

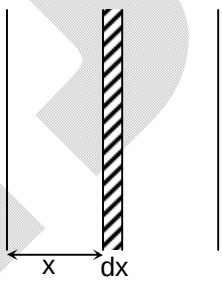
$$i = \frac{Bv}{\pi\lambda}$$

27. 3

Sol. $$dC = \frac{kL^2\varepsilon_0}{dx}$$

$$dC = \frac{B}{x^2}\frac{L^2\varepsilon_0}{dx}$$

$$\frac{1}{C_e} = \int\frac{1}{dC} \Rightarrow C_e = \frac{3BL^2\varepsilon_0}{d^3}$$



28. 3

Sol. $$mvr = n\frac{h}{2\pi} \quad \ldots(i)$$

$$\frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{3qq}{r^2} \quad \ldots(ii)$$

From (i) and (ii)

$$v = \frac{3q^2}{2\varepsilon_0 n.h}$$

29. 8

Sol. $$\frac{a_i + a_r}{a_i - a_r} = 4 \Rightarrow \frac{a_r}{a_i} = \frac{3}{5}$$

$$\frac{E_r}{E_i} = \left(\frac{a_r}{a_i}\right)^2 = \frac{9}{25}$$

Percentafge of energy reflected is

$\frac{E_r}{E_i} \times 100 = \frac{9}{25} \times 100 = 36\%$

$\therefore$ percentage of energy transmitted = 100 − 36 = 64

30. 3

Sol. $\beta_1 = \frac{D\lambda}{a}$

$\beta_2 = \frac{D\lambda}{a+2h}$

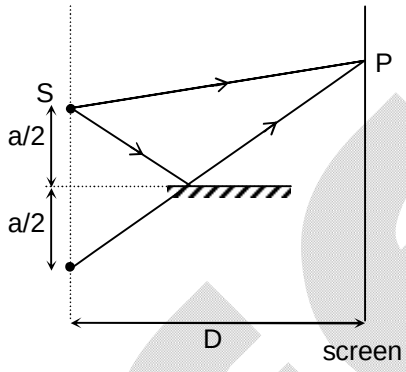
$\frac{\beta_1}{\beta_2} = \frac{a+2h}{a}$

$\Rightarrow \frac{\beta_1 - \beta_2}{\beta_2} = \frac{2h}{a}$

$a = 2h\frac{\beta_2}{(\beta_1 - \beta_2)} = 2.4 \text{ mm}$

$\beta_1 = \frac{D\lambda}{a} \Rightarrow \lambda = \frac{\beta_1 a}{D}$

$\lambda = 0.3 \times 10^{-6}$ m



Chemistry

PART – B

SECTION – A

31. C

Sol. $Ag_2CrO_4(s) \longrightarrow 2Ag^+(aq) + CrO_4^{2-}(aq)$

$E^o = 0.446 - 0.799 = -0.353\ V$

$\Delta G^o = -nFE^o = -2 \times 96500 \times (-0.353)$

$\Delta G^o = -2.303RT\log K_{SP} = 6.81 \times 10^4\ J/mole$

$6.81 \times 10^4 = -8.314 \times 298 \times 2.303 \log K_{SP}$

$K_{SP} = 1.16 \times 10^{-12}$

32. A

Sol. $K_a = \dfrac{[H^+][A^-]}{[HA]}$

$= \dfrac{(0.038 \times 0.1)^2}{(0.1 - 0.038 \times 0.1)} = 1.5 \times 10^{-4}$

% ionization of new concentration 'x' of acid is

$1.5 \times 10^{-4} = \dfrac{x^2}{0.5 - x}$

$x = \sqrt{1.5 \times 10^{-4} \times 0.5}$

= 0.0087

% ionization $= \dfrac{x}{0.5} \times 100 = 1.74\%$

33. C

Sol. $BaA_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2NaA$

$\frac{10g}{M}$	$\frac{8.93}{233.4}$
$\frac{10}{M}$	0.0382

$M = \dfrac{10}{0.0382} = 261.4$

34. B

Sol. Degree of dissociation increases.
Electrical conductivity increases.

35. C

Sol. Alkalie metal react with water vigorously.

36. A

37. A

Sol. $\ln[A]_t = -kt + \ln[A_o]$

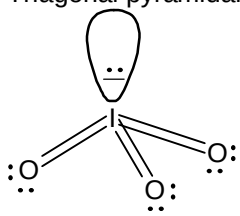
$$kt = \ln\frac{[A_o]}{[A]_t}$$

$$k = \frac{1}{t}\ln\frac{[A_o]}{[A]_t} = \frac{1}{300}\ln\frac{1}{0.99}$$

$$= 3.35\times10^{-5}\ S^{-1}$$

38. B

Sol. Triagonal pyramidal with 7 lone pair



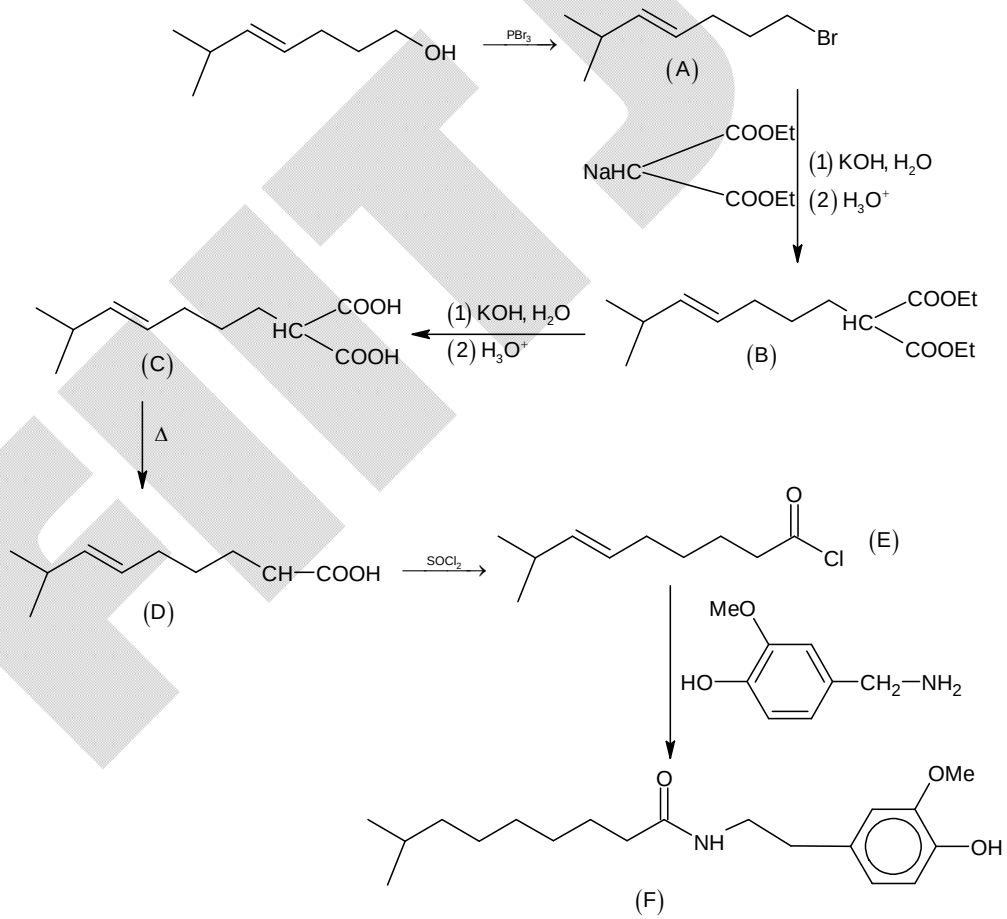
39. D

40. B

41. C

42. D

Sol.



43. C

Sol. $Ni^{2+} + 2e^- \longrightarrow Ni$

$2 \times 96500 \longrightarrow 58.699$

$58.69 \text{ gm} \longrightarrow 2 \times 96500$

$1.30 \longrightarrow \dfrac{2 \times 96500 \times 1.30}{58.69} = 4275.00$

$C = A \times Sec$

$sec = \dfrac{C}{A} = \dfrac{4275.00}{5} = 8.55 \times 10^2 \text{ S}$

44. A

45. A

46. D

Sol. $3Cu(s) + 2NO_3^-(aq) + 8H^+(aq) \longrightarrow 3Cu^{2+}(aq) + 2NO(g) + 4H_2O(\ell)$

47. C

48. C

Sol. 1.72 $\overset{o}{A}$

49. D

Sol. $h\nu = h\nu_o + KE$

$h\nu = h\nu_o + \dfrac{1}{2}mv^2$

$V = \sqrt{\dfrac{2hc}{m}\left(\dfrac{1}{\lambda} - \dfrac{1}{\lambda_o}\right)}$

50. A

Sol. The Clausius-Clapeyron equation best describe the phase transition between liquid and gas.

$\ell n P_{vap} = -\dfrac{\Delta H}{R}\left(\dfrac{1}{T}\right) + C$

We see that P_{vap} increases with T but not linearly.

$\therefore \ell n\dfrac{P_2}{P_1} = -\dfrac{\Delta H}{R}\left(\dfrac{1}{T_2} - \dfrac{1}{T_1}\right)$

$\therefore$ When T is lower, P_{vap} increases faster.

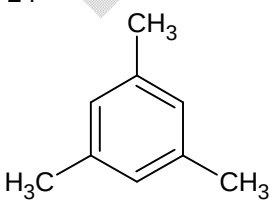
SECTION – B

51. 1

Sol. For P shell ℓ = 1

52. 24

Sol.

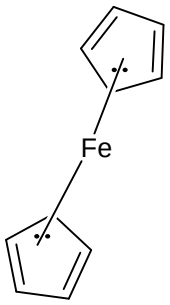


σ bonds = 21
π bonds = 3

53. 6

54. 12

Sol.



55. 6

Sol. $\text{Decapeptide} + 9H_2O = \text{Hydrolysed product} + 9H_2O$

$= 796 + 9 \times 18 = 958$ g

% of glycine = 47% of 958 = 450 gm

$\therefore$ Number of glycine unit $= \dfrac{450}{75} = 6$ unit

56. 2

Sol. $C = \dfrac{1}{50}; \dfrac{\ell}{a} = \dfrac{2.2}{4.4} = \dfrac{1}{2}$

$K = C \times \dfrac{\ell}{a} = \dfrac{1}{100}$

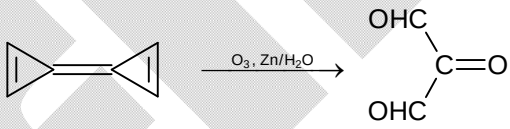
$\lambda_m = K \times \dfrac{1000}{M}$

$= 10^{-2} \times \dfrac{1000}{0.5} = 20 \text{ S cm}^2 / \text{mole}$

$= 2 \times 10^{-3} \text{ Sm}^2 / \text{mole}$

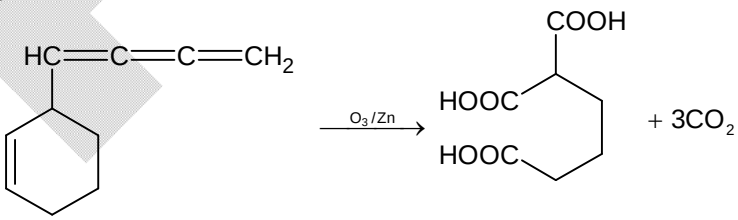
57. 5

Sol.



58. 3

Sol.



59. 1

60. 3

Mathematics

PART – C

SECTION – A

61. A

Sol. By the definition of scalar products and the law of cosines, we know

$$\overrightarrow{AB}\cdot\overrightarrow{AC} = cb\cos A = \frac{b^2+c^2-a^2}{2}.$$

In the same way,

$$\overrightarrow{BA}\cdot\overrightarrow{BC} = \frac{a^2+c^2-b^2}{2} \text{ and } \overrightarrow{CA}\cdot\overrightarrow{CB} = \frac{a^2+b^2-c^2}{2}$$

Then the original equation becomes

$$b^2+c^2-a^2+2\left(a^2+c^2-b^2\right) = 3\left(a^2+b^2-c^2\right),$$

or $a^2 + 2b^2 = 3c^2$. so we have

$$\cos C = \frac{a^2+b^2-c^2}{2ab} = \frac{a^2+b^2-\frac{1}{3}\left(a^2+2b^2\right)}{2ab}$$

$$= \frac{a}{3b}+\frac{b}{6a} \geq 2\sqrt{\frac{a}{3b}\cdot\frac{b}{6a}} = \frac{\sqrt{2}}{3}.$$

Then $\sin C = \sqrt{1-\cos^2 C} \leq \frac{\sqrt{7}}{3}$, and the equality holds if and only if $a:b:c = \sqrt{3}:\sqrt{6}:\sqrt{5}$.

Therefore the maximum value of sin C is $\frac{\sqrt{7}}{3}$.

62. D

Sol. Define $a_n = f\left(\frac{1}{n}\right)(n = 1,2,3...)$; then $a_1 = f(1) = 1$. For $x = -\frac{1}{k}\left(k \in N^+\right)$. Note that

$$\frac{x}{x-1} = \frac{-\frac{1}{k}}{-\frac{1}{k}-1} = \frac{1}{k+1}, \text{ and f(x) is an odd function.}$$

We have

$$f\left(\frac{1}{k+1}\right) = -\frac{1}{k}\cdot f\left(-\frac{1}{k}\right) = \frac{1}{k}\cdot f\left(\frac{1}{k}\right),$$

and that means $\frac{a_{k+1}}{a_k} = \frac{1}{k}$. Therefore,

$$a_n = a_1\cdot\prod_{k=1}^{n-1}\frac{a_{k+1}}{a_k} = \prod_{k=1}^{n-1}\frac{1}{k} = \frac{1}{(n-1)!}.$$

Then

$$\sum_{i=1}^{50} a_i a_{101-i} = \sum_{i=1}^{50}\frac{1}{(i-1)!\cdot(100-i)!} = \sum_{i=0}^{49}\frac{1}{i!\cdot(99-i)!}$$

$$= \frac{1}{99!}\cdot\sum_{i=0}^{49}{}^{99}C_i = \frac{1}{99!}\cdot\sum_{i=0}^{49}\frac{1}{2}\left({}^{99}C_i + {}^{99}C_{99-i}\right)$$

$$= \frac{1}{99!} \times \frac{1}{2} \times 2^{99} = \frac{2^{98}}{99!}.$$

63. A

Sol. Clearly, $f(x+14) = -\frac{1}{f(x+7)} = f(x)$, and it follows that

$$f(-100) = f(-100 + 14 \times 7) = f(-2) = -\frac{1}{f(5)} = -\frac{1}{\log_2 4} = -\frac{1}{2}.$$

64. B

Sol. Since

$x^2 = 1 - 2\cos y \in [-1, 3]$,

it follows that $x \in \left[-\sqrt{3}, \sqrt{3}\right]$. By $\cos y = \frac{1-x^2}{2}$, we have

$$x - \cos y = x - \frac{1-x^2}{2} = \frac{1}{2}(x+1)^2 - 1.$$

When $x = -1$, $x - \cos y$ attains the minimum value $-1\left(\text{as } y \text{ equals } \frac{\pi}{2}\right)$; when

$x = \sqrt{3}, x - \cos y$ attains the maximum value $\sqrt{3}+1$ (as y equals π). Since the range of

$\frac{1}{2}(x+1)^2 - 1$ is $\left[-1, \sqrt{3}+1\right]$, we find the range of $x - \cos y$ as $\left[-1, \sqrt{3}+1\right]$.

65. D

Sol. Let abc be a stable number.

If b = 0, then a = 1, c $\in$ (0, 1), and there are 2 stable numbers.

If b = 1, then a $\in$ {1, 2}, c $\in$ {0, 1, 2}, and there are $2 \times 3 = 6$ stable numbers.

If $2 \le b \le 8$, then $a, c \in \{b-1, b, b+1\}$, and there are $7 \times 3 \times 3 = 63$ stable numbers.

If b = 9, then a, c $\in$ {8, 9}, and there are $2 \times 2 = 4$ stable numbers.

Therefore, the number of stable is 2 + 6 + 63 + 4 = 75.

66. B

Sol. We have

$$7 + 4i = (z + \bar{w})(\bar{z} - w) = |z|^2 - |w|^2 - (zw - \overline{zw}).$$

Since $|z|^2$ and $|w|^2$ are real, and $\text{Re}(zw - \overline{zw}) = 0$, then

$|z|^2 - |w|^2 = 7$ and $zw - \overline{zw} = -4i$.

As $|z| = 3$, $|w|^2 = 2$. Then

$$(z + 2\bar{w})(\bar{z} - 2w) = |z|^2 - 4|w|^2 - 2(zw - \overline{zw})$$

$= 9 - 8 + 8i = 1 + 8i$.

67. B

Sol. Let $\log_u v = a$, $\log_v w = b$. Then

$$\log_v u = \frac{1}{a}, \log_w v = \frac{1}{b}, \log_u vw = \log_u v + \log_u v \cdot \log_v w = a + ab.$$

The given formulas become

$a + ab + b = 5$ and $\frac{1}{a}+\frac{1}{b}=3$,

from which we have $ab=\frac{5}{4}$. Therefore

$\log_w u = \log_w v \cdot \log_v u = \frac{1}{ab} = \frac{4}{5}$.

68. D

Sol. Suppose that Adam wrote down the coefficient a as a′ wrongly.

Then $-\frac{b}{a'}=6, \frac{c}{a'}=8$,

Therefore $-\frac{b}{c}=\frac{3}{4}$. Based on Ben's result, we have

$\left|\frac{b}{a}\right|=3, \left|\frac{c}{a}\right|=4$.

Since $\Delta = b^2 - 4ac < 0$, we have ac > 0, therefore $\frac{c}{a}=4$, and hence $\frac{b}{a}=-3$

Then,

$\frac{2b+3c}{a}=2(-3)+3(4)=6$.

69. C

Sol. Let $n = 10^{100}$. Then

$$\frac{10^{20000}}{10^{100}+3}=\frac{n^{200}}{n+3}=\frac{\left(n^2\right)^{100}-\left(3^2\right)^{100}}{n+3}+\frac{9^{100}}{n+3}$$

$$=\frac{\left(n^2-3^2\right)M}{n+3}+\frac{9^{100}}{n+3}=(n-3)M+\frac{9^{100}}{n+3}.$$

Since $9^{100} < n$, we have

$$\left[\frac{n^{200}}{n+3}\right]=(n-3)M=\frac{n^{200}-9^{100}}{n+3}=\frac{10^{20000}-81^{50}}{10^{100}+3}.$$

Since the units digit of $10^{20000} - 81^{50}$ is 9 and the units digit of $10^{100} + 3$ is 3, the units digit of the quotient must be 3.

70. A

Sol. Let a, b and c denote the number of –1s, 1s and 2s in the sequence, respectively. We need not consider the zeros. Then a, b, c are nonnegative integers satisfying

$– a + b + 2c = 19$ and $a + b + 4c = 99$.

It follows that $a = 40 – c$ and $b = 59 – 3c$, where $0 \le c \le 19$ (since $b \ge 0$),

So, $x_1^3 + x_2^3 + ... + x_n^3 = -a + b + 8c = 19 + 6c$.

When c = 0(a = 40, b = 59), the lower bound (19) is achieved.

When c = 19(a= 21, b = 2), the upper bound (133) is achieved.

71. B

Sol. According to the given condition, we have

$$f(x)=\begin{cases} x^2 & (x \ge 0) \\ -x^2 & (x<0) \end{cases}$$

So $2f(x) = f\left(\sqrt{2}x\right)$. Therefore, the original inequality is equivalent to $f(x+a) \geq f\left(\sqrt{2}x\right)$.

As f(x) is increasing over R, then $x + a \geq \sqrt{2}x$, i.e.,

$a \geq \left(\sqrt{2}-1\right)x$

Furhermore, since $x \in [a, a+2]$, $\left(\sqrt{2}-1\right)x$ reaches $\left(\sqrt{2}-1\right)(a+2)$ the maximum value when x – a+ 2.

Therefore, $a \geq \left(\sqrt{2}-1\right)(a+2)$, from which the obtain $a \geq \sqrt{2}$, i.e., $a \in [\sqrt{2}, +\infty)$.

The answer is then $[\sqrt{2}, +\infty)$

72. C

Sol. As sin x is a convex function for $x \in \left(0, \frac{\pi}{6}\right)$, we have $\frac{3}{\pi}x < \sin x < x$. Then

$$\sin\frac{\pi}{13} < \frac{\pi}{13} < \frac{1}{4},\ \sin\frac{\pi}{12} > \frac{3}{\pi}\times\frac{\pi}{12} = \frac{1}{4}$$

$$\sin\frac{\pi}{10} < \frac{\pi}{10} < \frac{1}{3}, \sin\frac{\pi}{9} > \frac{3}{\pi}\times\frac{\pi}{9} = \frac{1}{3},$$

That is

$$\sin\frac{\pi}{13} < \frac{1}{4} < \sin\frac{\pi}{12} < \sin\frac{\pi}{11} < \sin\frac{\pi}{10} < \frac{1}{3} < \sin\frac{\pi}{9}.$$

Therefore, all the possible values of positive integers n are 10 . 11. 12 and their sum is 33.

73. A

Sol. $$\lim_{x\to 1}\frac{\sqrt[3]{x^2+7}-\sqrt{x+3}}{x^2-3x+2}$$

$$= \lim_{x\to 1}\frac{\sqrt[3]{x^2+7}-2}{x^2-3x+2} + \lim_{x\to 1}\frac{2-\sqrt{x+3}}{x^2-3x+2}$$

$$= \lim_{x\to 1}\frac{x+1}{(x-2)\left(\sqrt[3]{(x^2+7)^2}+2\sqrt[3]{x^2+7}+4\right)} + \lim_{x\to 1}\frac{1}{(2-x)\left(2+\sqrt{x+3}\right)}$$

$$= -\frac{2}{12}+\frac{1}{4} = \frac{1}{12}$$

74. D

Sol. $A^T - I = A^TA$

take transpose on both the sides

$\Rightarrow$ $A - I = A^TA \Rightarrow A^T - I = A - I$

$\Rightarrow$ $A^T = A \Rightarrow$ A is symmetric

Now $A - I = A^2 \Rightarrow A^2 - A + I = 0$

75. A

Sol. Let $I = \int\frac{3+2\cos x}{(2+3\cos x)^2}dx$

Multiplying Nr. and Dr. by cosec^2x

$\Rightarrow$ $I = \int\frac{(3\text{cosec}^2x + 2\cot x\,\text{cosec}x)}{(2\text{cosec}x+3\cot x)^2}dx$

$$= -\int \frac{(-3\text{cosec}^2 x - 2\cot x \text{cosec} x)}{(2\text{cosec} x + 3\cot x)^2} dx$$

$$= \frac{1}{2\text{cosec} x + 3\cot x} = \left(\frac{\sin x}{2 + 3\cos x}\right) + c$$

76. D

Sol. For (a, b), (c, d) $\in N \times N$

(a, b) R (c, d) $\Rightarrow$ ad (b + c) = bc (a + d)

Reflexive : Since, ab (b + a) = ba(a + b) $\forall$ a, b $\in$ N

$\therefore$ (a, b) R(a, b), $\therefore$ R is reflexive

Symmetric : For (a, b), (c, d) $\in N \times N$

Let (a, b) R(c, d)

$\therefore$ ad(b + c) = bc(a + d)

$\Rightarrow$ bc(a + d) = ad(b + c)

$\Rightarrow$ cb(d + a) = da(c + b)

$\Rightarrow$ (c, d) R(a, b) $\therefore$ R is symmetric

Transitive : For (a, b), (c, d), (e, f) $\in N \times N$

Let (a, b) R(c, d), (c, d) R(e, f) and ad(b + c) = bc(a + d), cf(d + e) = de(c + f)

$\Rightarrow$ adb + adc = bca + bcd ….(i)

and cfd + cfe = dec + def …(ii)

(i) $\times$ ef + (ii) $\times$ ab gives,

adbef + adcef + cfdab + cfeab = bcaef + bcdef + decab + defab

$\Rightarrow$ adcf(b + e) = bcde(a + f)

$\Rightarrow$ af(b + e) = be(a + f)

$\Rightarrow$ (a, b) R (e, f)

$\therefore$ R is transitive. Hence, R is an equivalence relation.

77. D

Sol. Since variance is independent of change of origin, hence the variance of the observations 101, 102, ….. , 200 is the same as the variance of the observation 151, 15, ……. , 50

$\therefore$ $V_A = V_B \Rightarrow \frac{V_A}{V_B} = 1$

78. D

Sol. $\alpha = \frac{\pi}{3}$

$\therefore$ $\frac{\cos 2\alpha + \sec\alpha + 3\sqrt{3}}{\tan\alpha} = \frac{-\frac{1}{2} + 2 + 3\sqrt{3}}{\sqrt{3}} = \frac{\frac{3}{2} + 3\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{3}}{2} + 3$

79. A

Sol. Let $B = \int_0^{\pi} \frac{2x^2 \cdot \cos^2 x/2}{(1+\sin x)^2} dx$

$\therefore$ $B - A = \int_0^{\pi} \frac{x^2 (2\cos^2 x/2)}{(1+\sin x)^2} dx - \int_0^{\pi} \frac{x^2 \cdot \cos x}{(1+\sin x)^2} dx$

Using by parts,

$$B - A = \left\{-\frac{x^2}{1+\sin x}\right\}_0^{\pi} + 2\int_0^{\pi} \frac{x}{(1+\sin x)} dx$$

$B - A = -\pi^2 + 2K$ …(i)

where, $K = \int_0^{\pi} \frac{x dx}{1+\sin x}$ $\left[\text{using } \int_0^{2a} f(x)dx = \begin{cases} 0, f(2a-x) = -f(x) \\ 2\int_0^a f(x)dx, f(2a-x) = f(x) \end{cases}\right.$

$\Rightarrow \quad 2K = 2\pi \int_0^{\pi/2} \frac{dx}{1+\sin x} = 2\pi \int_0^{\pi/2} \frac{dx}{1+\cos x}$

$= = 2\pi \int_0^{\pi/2} \frac{1}{2}\sec^2 \frac{x}{2} dx = 2\pi \left\{\tan \frac{x}{2}\right\}_0^{\pi/2} = 2\pi$

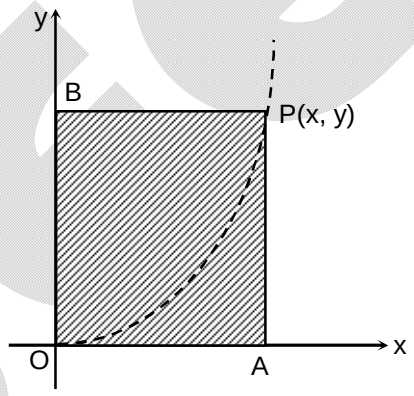
$\therefore \quad K = \pi$

Thus, $B - A - \pi^2 + 2K$

$\Rightarrow \quad B = A - \pi^2 + 2\pi$

80. A

Sol. $\frac{\text{Area of OBPO}}{\text{Area of OPAO}} = \frac{m}{n}$



$\Rightarrow \quad \frac{xy - \int_0^x ydx}{\int_0^x ydx} = \frac{m}{n} \Rightarrow nxy = (m+n)\int_0^x ydx$

Differentiating w.r.t. x, we get

$n\left(x\frac{dy}{dx} + y\right) = (m+n)y$

$\Rightarrow \quad nx\frac{dy}{dx} = my$

$\Rightarrow \quad \frac{m}{n} \cdot \frac{dx}{x} = \frac{dy}{y}$, integrating both the sides

$y = Cx^{m/n}$

SECTION – B

81. 16

Sol. We have

$f(x) = \left(\sin^2 \frac{kx}{10} + \cos^2 \frac{kx}{10}\right)^2 - 2\sin^2 \frac{kx}{10}\cos^2 \frac{kx}{10}$

$= 1 - \frac{1}{2}\sin^2 \frac{kx}{5} = \frac{1}{4}\cos \frac{2kx}{5} + \frac{3}{4}.$

f(x) reaches the maximum if and only if $x = \frac{5m\pi}{k}(m \in Z)$. Since any open interval (a, a + 1) with length 1 contains at least one maximum point, so $k > 5\pi$.

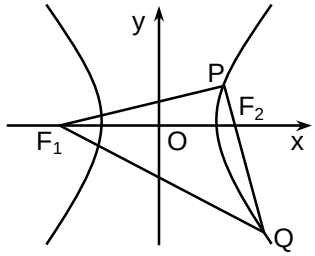
One the other hand, when $k > 5\pi$, any open interval (a, a + 1) will contains an entire period of f(x). Therefore, the minimum of k is $[5\pi] + 1 = 16$.

82. 7

Sol. By the property of a hyperbola, we know

$F_1F_2 = 2 \times \sqrt{1+3} = 4$ and $PF_1 - PF_2 = QF_1 - QF_2 = 2$.

Since $\angle F_1PQ = 90^o$ then $PF_1^2 + PF_2^2 = F_1F_2^2$. We have



$$PF_1 + PF_2 = \sqrt{2\left(PF_1^2 + PF_2^2\right) - \left(PF_1 - PF_2\right)^2}$$

$$= \sqrt{2\times 4^2 - 2^2} = 2\sqrt{7}\,.$$

Therefore, the inscribed circle radius of ΔF_1PQ is

$$r = \frac{1}{2}\left(F_1P + PQ - F_1Q\right) = \frac{1}{2}\left(PF_1 + PF_2\right) - \frac{1}{2}\left(QF_1 - QF_2\right)$$

$$= \sqrt{7} - 1.$$

83. 11

Sol. Clearly, we have A(3, 0) and F(0, 1). Let the coordinates of P be $\left(3\cos\theta, \sqrt{10}\sin\theta\right), \theta \in \left(0, \frac{\pi}{2}\right)$.

Then

$$S_{OAPF} = S_{\Delta OAP} + S_{\Delta OFP} = \frac{1}{2}\cdot 3\cdot\sqrt{10}\sin\theta + \frac{1}{2}\cdot 1\cdot 3\cos\theta$$

$$= \frac{3}{2}\left(\sqrt{10}\sin\theta + \cos\theta\right) = \frac{3\sqrt{11}}{2}\sin\left(\theta + \varphi\right),$$

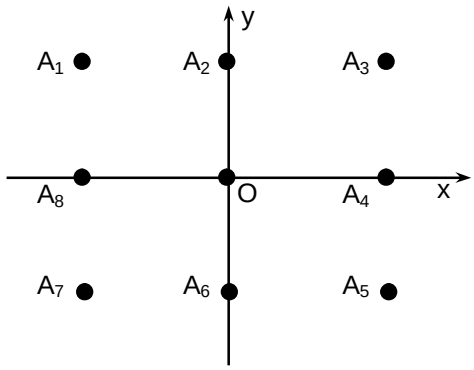
In which $\varphi = \tan^{-1}\frac{\sqrt{10}}{10}$.

When $\theta = \tan^{-1}\sqrt{10}$, quadrilateral OAPF has the maximal area as $\frac{3\sqrt{11}}{2}$.

84. 7

Sol. There are 9 points in K, and the number of ways of choosing 3 points from K is ${}^9C_3 = 84$.

Let the points be A_1, A_2, …, A_8 and O, as shown in the figure below.



There are 8 pairs of points at distance $\sqrt{5}$. Let us choose one pair among them. By symmetry, we may assume the chosen pair are A_1 and A_4. Since there are 7 ways to choose the other point, there are $7 \times 8 = 56$ unordered triples. On the other hand, each $A_i = (i = 1,2,....,8)$ is at distance $\sqrt{5}$ from points A_{i+3}, A_{i+5} (indices modulo 8). This implies that there are 8 triples

$$\left\{A_i, A_{i+3}, A_{i+5}\right\}(i = 1,2,....,8)$$

Being counted twice. Hence the total number of unordered triples is 56 – 8 = 48, and the probability is $\frac{48}{84} = \frac{4}{7}$.

85. 7

Sol. By definition of odd and even functions, we have

$f(1) + 1 = -[f(-1) + (-1)^2] = -f(-1) - 1$,

$f(1) + 2 = f(-1) + 1/2$

Adding the above equations to cancel f(–1), it following $2f(1) + 3 = -\frac{1}{2}$, or $f(1) = -\frac{7}{4}$.

86. 80

Sol. Assume that AB is on the line y = 2x – 17 and the coordinates of the other two vertices on the parabola are $C(x_1, y_1)$ and $D(x_2, y_2)$. Then CD is on a line L whose equation is y = 2x + b. Combining this with the equation of the parabola, we get $x^2 = 2x + b \Rightarrow x_{1,2} = 1 \pm \sqrt{b+1}$. Assume that the length of one side of the square is a. Then

$$a^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$5(x_1 - x_2)^2 = 20(b+1) \quad \ldots \text{(i)}$$

Pick a point (6, – 5) on the line y = 2x – 17, and the distance from the point to the line y = 2x + b is a.

So $a = \dfrac{|17 + b|}{\sqrt{5}}$ … (ii)

From (i) and (ii), we get $b_1 = 3$, $b_2 = 63$, so $a^2 = 80$ or $a^2 = 1280$, and $a^2_{min} = 80$.

87. 3

Sol. The coefficients of the first three terms are $1, \dfrac{1}{2}{}^nC_1$ and $\dfrac{1}{2^2}{}^nC_2$. By the assumption, we have

$$2 \cdot \frac{1}{2}{}^nC_1 = 1 + \frac{1}{2^2}{}^nC_2.$$

That is $n = 1 + \dfrac{1}{8}n(n-1)$.

Solving for n, we obtain n = 8 and n = 1 (not admissible).

When n = 8, $T_{r+1} = {}^8C_r \left(\dfrac{1}{2}\right)^r x^{\frac{16-3r}{4}}$, r = 0, 1, 2, …, 8. But r are 3 terms in the expansion of x with integer power.

88. 29

Sol. $2^{-x^2} < 2^{-x^2}\left[2^{x^2}\right] \le 1 \Rightarrow \left[2^{-x^2}\left[2^{x^2}\right]\right] = 0$ and whenever

$$2^{-x^2}\left[2^{x^2}\right] = 1$$

$$\left[2^{x^2}\right] = 2^{x^2}$$

Also if $x \in (-2, 2)$, then $1 \le 2^{x^2} < 16$

89. 12

Sol. Let sinx = t. The expression is then changed to $g(t) = (-at^2 + a - 3)t$, or
$g(t) = -at^3 + (a - 3)t$,
from $-at^3 + (a - 3)t \ge -3$, we get
$-at(t^2 - 1) - 3(t - 1) \ge 0$,
$(t - 1)(-at(t + 1) - 3) \ge 0$,
Since $t - 1 \le 0$, we have $-at(t + 1) - 3 \le 0$, or
$a(t^2 + t) \ge -3$ … (i)
When t = 0, – 1, expression (i) always holds, when $0 < t \le 1$, we have $0 < t^2 + t \le 2$, and when $-1 < t < 0$, $-\dfrac{1}{4} \le t^2 + t < 0$. Therefore $-\dfrac{3}{2} \le a \le 12$.

90. 4

Sol. Solving C_1 and C_2 we get P as $\left(\frac{r^2}{2}, \sqrt{r^2 - \frac{r^4}{4}}\right)$

Solving the line PQ with x-axis we get x co-ordinate as $\alpha = \frac{\frac{r^3}{2}}{r - \sqrt{r^2 - \frac{r^4}{4}}}$

$\Rightarrow \alpha_{lim} = \lim\limits_{r \to 0^+} \frac{\frac{r^3}{2}}{r - \sqrt{r^2 - \frac{r^4}{4}}} = 4$

$\Rightarrow$ R approaches the point (4, 0) as $r \to 0^+$